DATA SCIENCE SUMMIT · AI EDITION · ONLINE

The Scorecard Illusion

Why SHAP values are not points — and how they can fool you

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This talk is about **classic, tabular ML** — not LLMs.

The “boring” models that still decide what happens to your money.

The setup: a PD model

Probability of default — will this customer stop repaying within the next 12 months? It's fed by the customer's behavioural and credit history: delinquency record, max delay (worst payment delay in a look-back window), overdue amounts, utilization, age.

Logistic scorecard

Readable, regulator-friendly — but loses on performance.

XGBoost / LightGBM

Higher predictive power, catches hidden risk — but it's a black box.

Why do we even need to explain it

Hundreds of trees voting. Nobody reads that off the cuff. Yet we *must* say why a customer was declined — for three audiences:

Regulator

Prove the model isn't doing something stupid or discriminatory.

Business

Why did the model suddenly mass-reject this segment?

Customer

A *specific* reason for a decline — increasingly a legal right, not a courtesy.

A black box you are obliged to explain. **Enter SHAP.**

EU AI Act 2024/1689 (Annex III 5b: credit scoring high-risk; Art. 86) · PL ustawa o kredycie konsumenckim, art. 9–10.

What SHAP is, in one breath

From **cooperative game theory**: split a payout among players, fairly, in proportion to contribution.

The rules it obeys

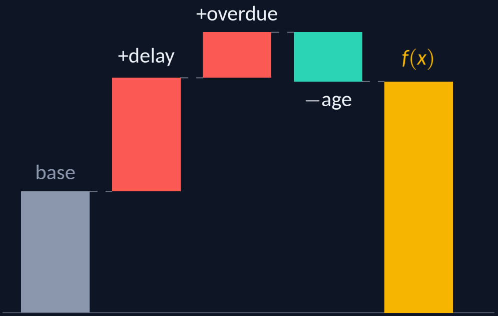
- No contribution $\Rightarrow$ zero
- Equal contribution $\Rightarrow$ equal share
- Shares sum to the whole payout

Mapped to our model

- Game $=$ the prediction
- Players $=$ the features
- Payout $=$ prediction $-$ average

Additivity: base value $+$ contributions $=$ *exactly* this customer's prediction.

One customer, decomposed — and why it's great



Local explanations. Global summaries when averaged. Fast for trees.

If I had to pick one explainer for tabular ML, I'd pick SHAP.

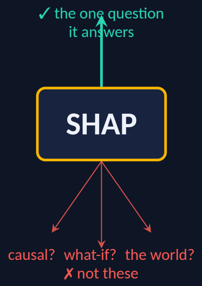
So why throw mud at it for 20 minutes?

Because it's so convenient that people trust it on questions it never answers.

SHAP doesn’t answer every question you have for your model

You want to ask your model many things: *why this score? what if I change X? which driver really matters? what’s actually going on?*

SHAP answers **one specific kind** of those questions well. Treat it as the answer to *all* of them and it quietly misleads you.



The thread for the rest of this talk: **SHAP explains the model, not the world.**

PITFALL #1

SHAP is not a linear model — and not a “what-if”

The single most common mistake I see.

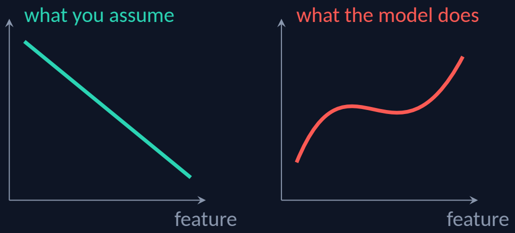
The tempting (wrong) logic

"SHAP gives additive contributions $\Rightarrow$ it's a local linear model $\Rightarrow$ I can tell the customer:"

> "Your *limit utilization* contributes $+5$ risk points — lower it and your risk drops."

SHAP does **not** answer "what happens if I change this feature." It answers "how do I split the gap vs. the background."

Change it for real $\rightarrow$ different tree paths, rebuilt interactions, a move SHAP never hinted at.



The credit example

A customer with **high limit utilization**, but **long tenure** and many products closed on time.

SHAP shows utilization with a *small* contribution — in interaction with tenure, the model reads it as “normal mode of a trusted customer.”

They take the letter literally: pay the limit to zero, close half their products.

They fall out of the “safe” interaction. PD doesn’t improve — sometimes it gets *worse*. SHAP never simulated that move; it only decomposed the old prediction.

The remedy

There is hard, peer-reviewed proof that complete-and-linear attribution methods (SHAP included) can do **no better than a coin flip** on tasks like algorithmic recourse.

For “what should the customer change,” use **counterfactual explanations** — they *simulate* the model to find the smallest decision-flipping change.

SHAP tells you **why** the model gave this score. It does *not* tell you **what to change** to get a different one — for that, use counterfactuals.

Bilodeau et al., *PNAS* 2024 (impossibility theorems) · Albini et al., *FAccT* 2022 (Counterfactual SHAP).

PITFALL #2

Don’t subtract a customer’s SHAP day over day

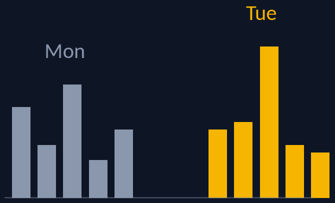
The trap that falls straight out of monitoring.

The reflex

Monday: PD $= 3\%$. Tuesday: PD $= 4\%$. The prediction moved.

Analyst's instinct: *compute SHAP both days, subtract, and the feature that moved is "the cause."*

> The change in SHAP attribution does **not** necessarily match the actual cause of the prediction change.



1 input changed $\longrightarrow$ 5 bars moved

Two reasons it breaks

1 · Interactions

One feature changes (e.g. a new delinquency episode) and it shifts the attributions of features that *didn't* change. You see "five features moved" when one thing changed.

2 · Background

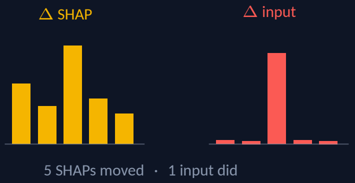
SHAP is always measured against a *reference set* of customers — the base value on that waterfall *is* this background. Swap it or let it drift and every SHAP moves, even with identical inputs.

Engineering trap: `TreeExplainer` with no background **silently** picks *path-dependent* over *interventional*. Different numbers — and you may not know which mode you're in.

TreeSHAP modes: Lundberg et al., *NMI* 2020 · consequence: Janzing et al. 2020; Sundararajan & Najmi 2020.

So: watch the inputs, not just the SHAP

It’s fine to watch SHAP move over time — but know that changing *one* feature reshuffles the SHAP of *other*, unchanged features (interactions). The bar that moved may not be the feature that changed.



Don’t diagnose a moved prediction from SHAP alone. Track how the **inputs** moved too, and read the two together.

PITFALL #3

“Feature importance” has two faces

Average |SHAP| over the population — and both faces hurt.

Face 1 — correlation steals the crown

A **family** of features for one phenomenon: max delay at 3/6/9/12 months, overdue now / max / average.

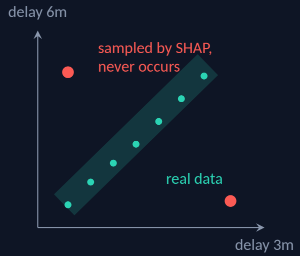
Next to it, one **lonely** feature: *age*.

The delinquency contribution splits across five bars; age keeps its whole bar.

Age can top the ranking — not because it matters most, but because it has **no one to share with.** A correlation artifact, not truth about the model.

Face 1 — and it gets worse

With strong correlation, SHAP samples feature **combinations that don't exist** — max delay $= 90$ days in one window, 0 in the next — and scores the model on those *unrealistic* points.



So part of the attribution is built on data the model will never actually see.

Correlated features & unrealistic samples: Aas, Jullum & Løland, *Artificial Intelligence* 298:103502 (2021).

Face 1 — why: SHAP builds rows that can't exist

Two correlated features — **max delay over 3m** and **over 6m** — where by construction $6\text{m} \geq 3\text{m}$. To score "delay 3m" alone, SHAP keeps it and draws "delay 6m" from the background:

Real customer

delay 3m $= 90$, delay 6m $= 95$

✓ plausible

Row SHAP scores

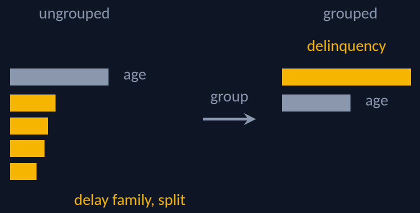
delay 3m $= 90$, delay 6m $= 0$

✗ impossible ($6\text{m} < 3\text{m}$)

The model is queried on that impossible row — and what it predicts there leaks into the explanation.

Face 1 — the remedy

Group correlated features. Group SHAP values can be summed — so compare the *delinquency phenomenon* against age, not one-of-five delay features against age.



Compare *phenomena*, not individual correlated columns — then the ranking reflects the model, not the feature layout.

Face 2 — importance is computed on *today's* data

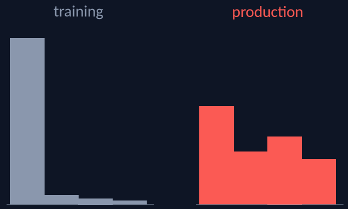
Take a normally-quiet signal — e.g. **# applications sharing one phone number across different PESELs**. In training it's almost always zero, so by mean $|\text{SHAP}|$ it ranks among the **least important** features.

99.5%

zeros in training

90%

zeros in production



Then operators start **recycling** phone numbers — the same number legitimately reappears across people. The feature quietly goes from 99.5% to 90% zeros and turns into a real driver — but it sat at the bottom of the ranking, so nobody was watching.

Face 2 — the remedy

A rare feature can be *weak on average but decisive when it fires*: low mean $|\text{SHAP}|$, high **maximum** $|\text{SHAP}|$.

Rank and monitor features by the **max** (or the tail) of $|\text{SHAP}|$, not only the mean — that's where rare-but-powerful drivers hide.

A distinction to carry

SHAP importance $\neq$ permutation importance.

Permutation

Overfit on a noise feature? Says *zero* — it adds nothing to quality.

SHAP

Same feature? Says *important* — it moves the output, regardless of quality.

Two definitions, not a bug — so decide what you need importance *for*, then pick the method.

Problems with Shapley as importance: Kumar et al., ICML 2020 · “Many Shapley values”: Sundararajan & Najmi, ICML 2020.

PITFALL #4

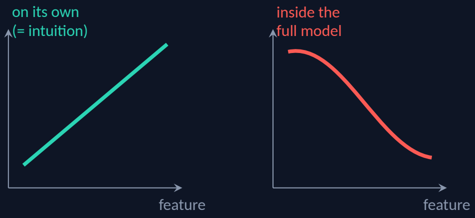
A feature in the model is not a model of that feature

Especially with correlated features — and rarely what intuition expects.

Alone vs. inside the model

Build a model on *one* feature — say short delinquency episodes — and it behaves as intuition says: more episodes, more risk.

Inside the **full** model, correlated with "always closed on time" and other signals, the *same* feature can push the *opposite* way.



A feature's effect in the full model is not its effect alone — and not what intuition predicts. Read its SHAP in isolation and your business conclusion can come out backwards.

The one rule

SHAP explains the model, not the world.

Want conclusions about the *world*? You need causal assumptions — a deliberate choice, not a default `shap_values()` call.

Causal variants: Heskes et al., NeurIPS 2020; Janzing, Minorics & Blöbaum, AISTATS 2020.

Four pitfalls, one sentence each

- Not a what-if analysis.
- Don’t read causes into day-to-day SHAP — watch the inputs too.
- Importance is an artifact of correlation *and* the current distribution.
- A feature’s effect in the model isn’t its effect alone — or what intuition says.

And it can be **deliberately fooled** — a discriminatory model can show SHAP innocent-looking explanations — and it’s unstable under small input changes.

Fooling SHAP: Slack, Hilgard, Jia, Singh & Lakkaraju, *AIES* 2020.

But don't bin SHAP — ask it the right question

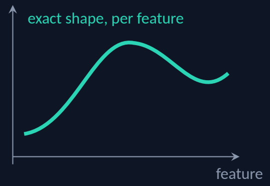
I use it every day. It's the best post-hoc explainer we have for tabular data.

Great when you ask: *"how did this model decompose this prediction vs. the background?"*

Dangerous when you ask it causal, counterfactual, or about-the-world questions.

When stakes are high — consider a glass box

Instead of a black box patched with SHAP, build a model transparent *by nature*: the **Explainable Boosting Machine** (EBM) — a GAM with a drawn, exact shape per feature.



0.975
EBM (AUROC)

0.981
XGBoost

Competitive, not "better" — trade one AUROC point for full transparency.

EBM benchmark (default params): Nori et al., "InterpretML", arXiv:1909.09223 · Rudin, *NMI* 2019.

Love SHAP.
Ask only what it answers.

When the stakes are high — consider a glass box.

Thank you. Questions?

References (slide footnotes, consolidated)

Regulatory (the case for explanation)

EU AI Act, Reg. (EU) 2024/1689 — credit scoring high-risk (Annex III, 5(b)); Art. 86 right to explanation.

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Pitfall 1 — not a what-if

Bilodeau, Jaques, Koh, Kim — Impossibility theorems for feature attribution. *PNAS* 121(2) e2304406120 (2024).

Albini, Long, Dervovic, Magazzeni — Counterfactual Shapley Additive Explanations. *FAccT* '22, 1054–1070.

Pitfall 2 — day over day

Lundberg et al. — From local explanations to global understanding…*Nat. Mach. Intell.* 2(1):56–67 (2020).

Pitfalls 3 & 4 — importance & interactions

Aas, Jullum, Løland — Explaining individual predictions when features are dependent. *Artif. Intell.* 298:103502 (2021).

Kumar, Venkatasubramanian, Scheidegger, Friedler — Problems with Shapley-value-based explanations. *ICML* 2020.

Sundararajan & Najmi — The Many Shapley Values. *ICML* 2020.

Heskes et al. — Causal Shapley Values. *NeurIPS* 2020.

Janzing, Minorics, Blöbaum — Feature relevance quantification. *AISTATS* 2020.

Close — robustness & glass box

Slack et al. — Fooling LIME and SHAP. *AIES* 2020.

Nori, Jenkins, Koch, Caruana — InterpretML. arXiv:1909.09223.

Rudin — Stop explaining black box models…*Nat. Mach. Intell.* 2019.

Slides & references

github.com/wlazlod/conference-talks

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